



INTRODUCTION

Small Unmanned Aerial Vehicles (UAVs) and fixed-wing drones operate in a highly complex aerodynamic regime ($Re \approx 400,000$) where laminar-turbulent transition significantly impacts performance. Traditional gradient-based methods often struggle with the non-linear, multi-modal nature of aerodynamic objective functions.

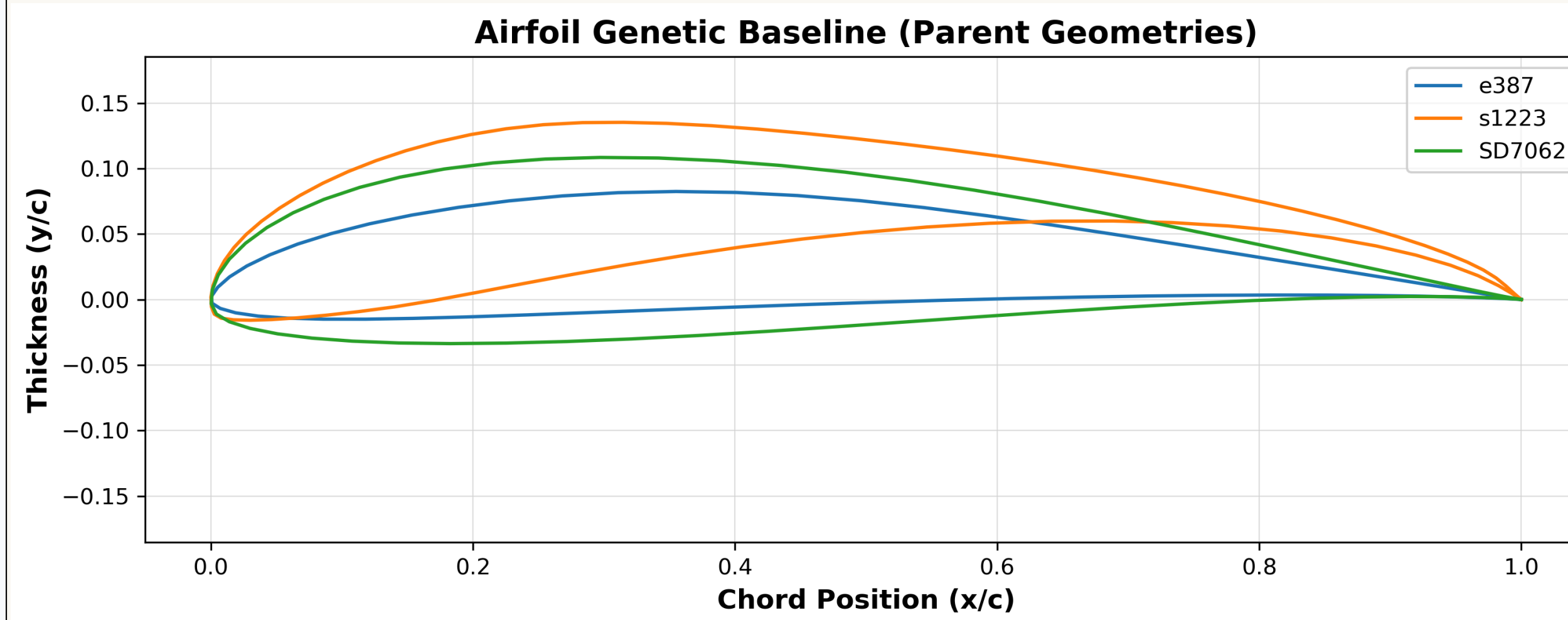


Figure 1. The initial genetic pool consisting of three diverse low-Reynolds parent geometries, providing a rich 2D design space for the multi-objective NSGA-II algorithm.

This research employs **Non-dominated Sorting Genetic Algorithm II (NSGA-II)** with **Class-Shape Transformation (CST)** parameterization utilizing three diverse parent airfoils. This creates an exponentially richer 2D design space, enabling the genetic algorithm to discover genuinely novel, highly efficient geometries that lie strictly outside the convex hull of the parent designs.

OBJECTIVES

- Maximise Lift-to-Drag ratio (L/D) at cruise condition $\alpha = 6^\circ$, $Re = 4 \times 10^5$
- Maximise peak lift coefficient (Cl_{max}) across full polar sweep ($\alpha = 0^\circ - 10^\circ$)
- Generate a Pareto-optimal set of airfoils capturing the full L/D vs. Cl_{max} trade-off
- Enforce structural feasibility via minimum thickness-to-chord constraint ($t/c \geq 0.09$)

KEY HIGHLIGHTS

- ♦ Peak L/D improved by **+33.8%**
- ♦ **15** Pareto-optimal solutions generated
- ♦ Convergence validated over **2** independent runs

METHODOLOGY

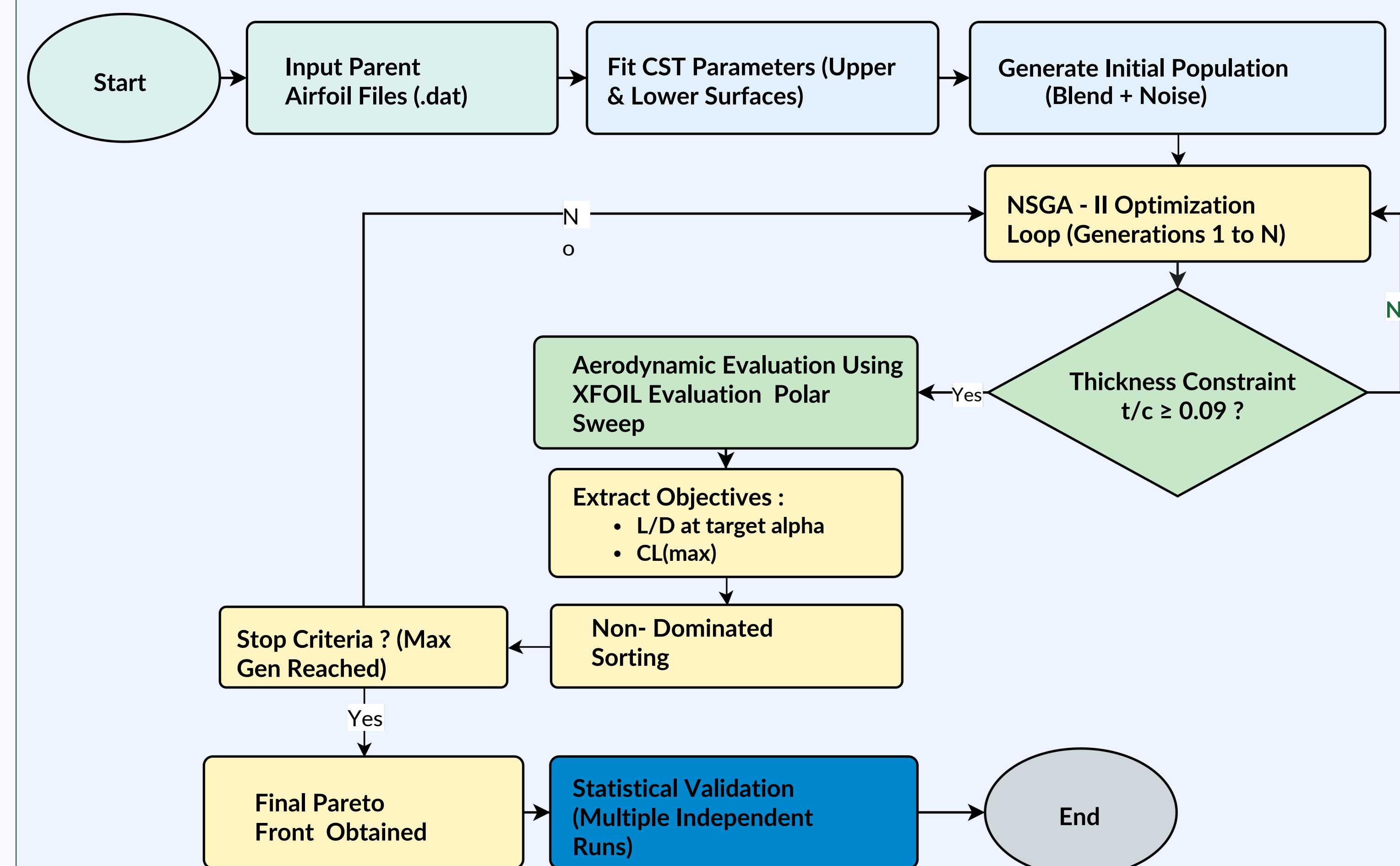


Figure 2. Algorithmic Framework. Flowchart detailing the integration of XFOIL aerodynamic evaluation with CST geometry parameterization and the NSGA-II evolutionary loop.

RESULTS

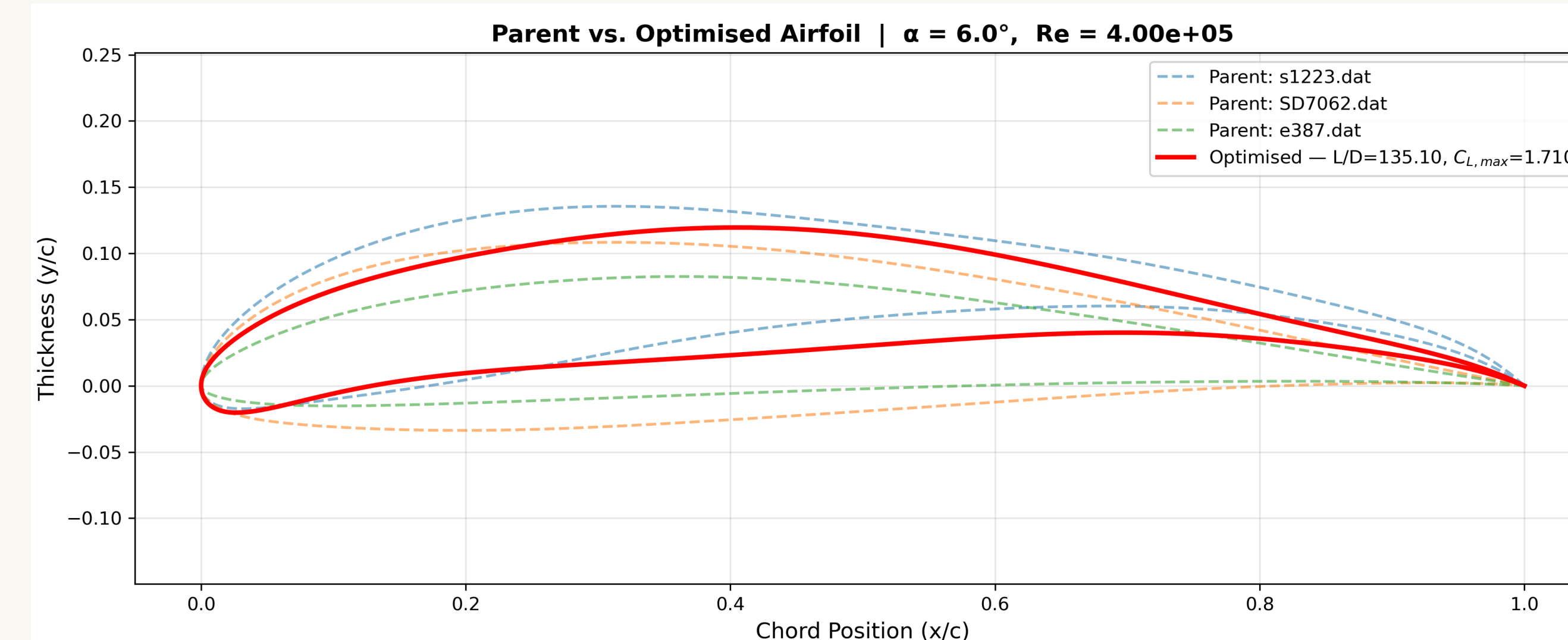


Figure 3. Geometric Evolution. Physical overlay of the optimized CST geometry against the baseline parents. The NSGA-II optimised airfoil features a higher-cambered, aerodynamically refined profile that outperforms all three parent airfoils at the design condition $\alpha = 6^\circ$, $Re = 4 \times 10^5$

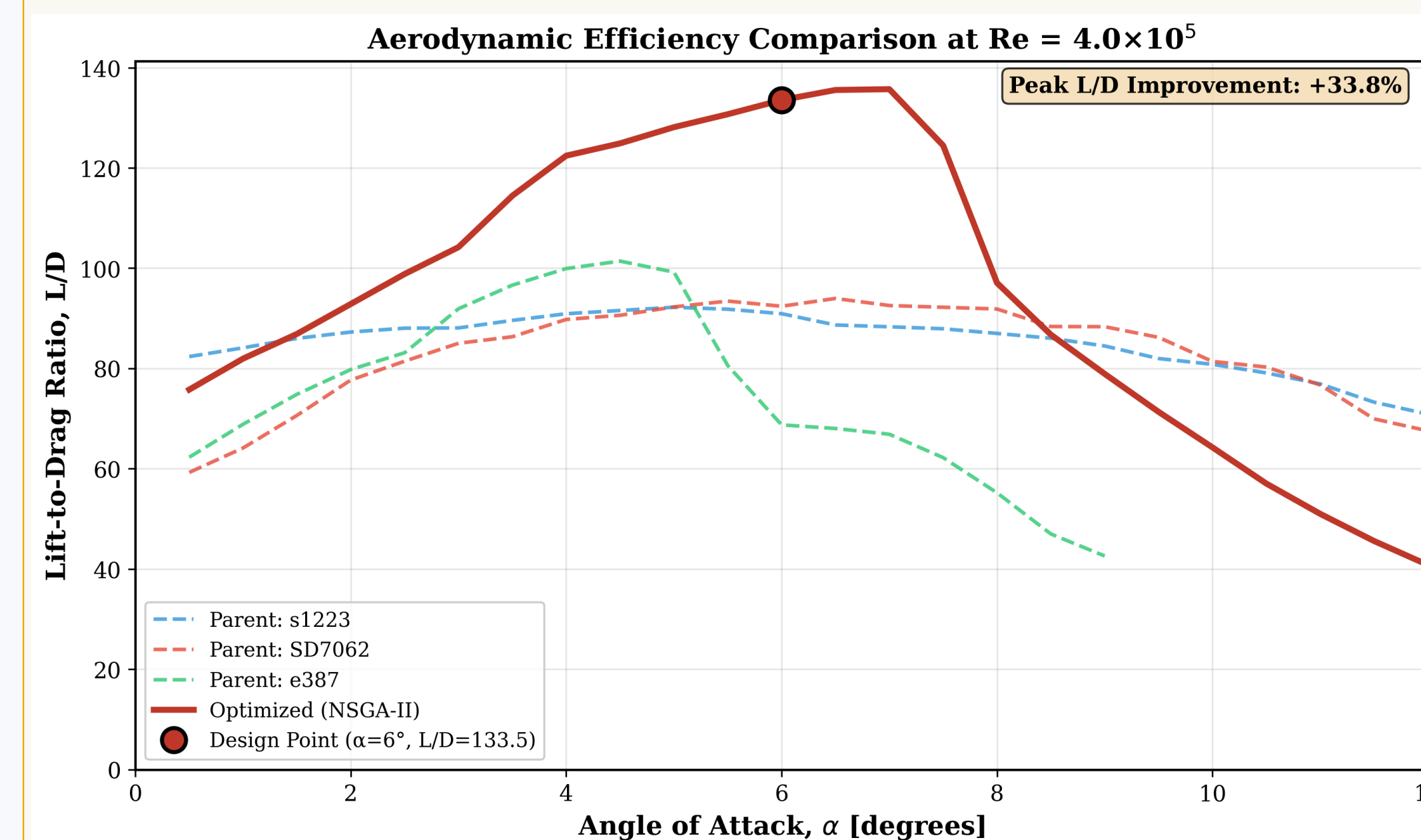


Figure 4. Aerodynamic Efficiency comparison. Optimized profile demonstrates a significant improvement over the best-performing parent baseline (E387) at the design point, while maintaining robust off-design performance across the operational envelope

PERFORMANCE ANALYSIS

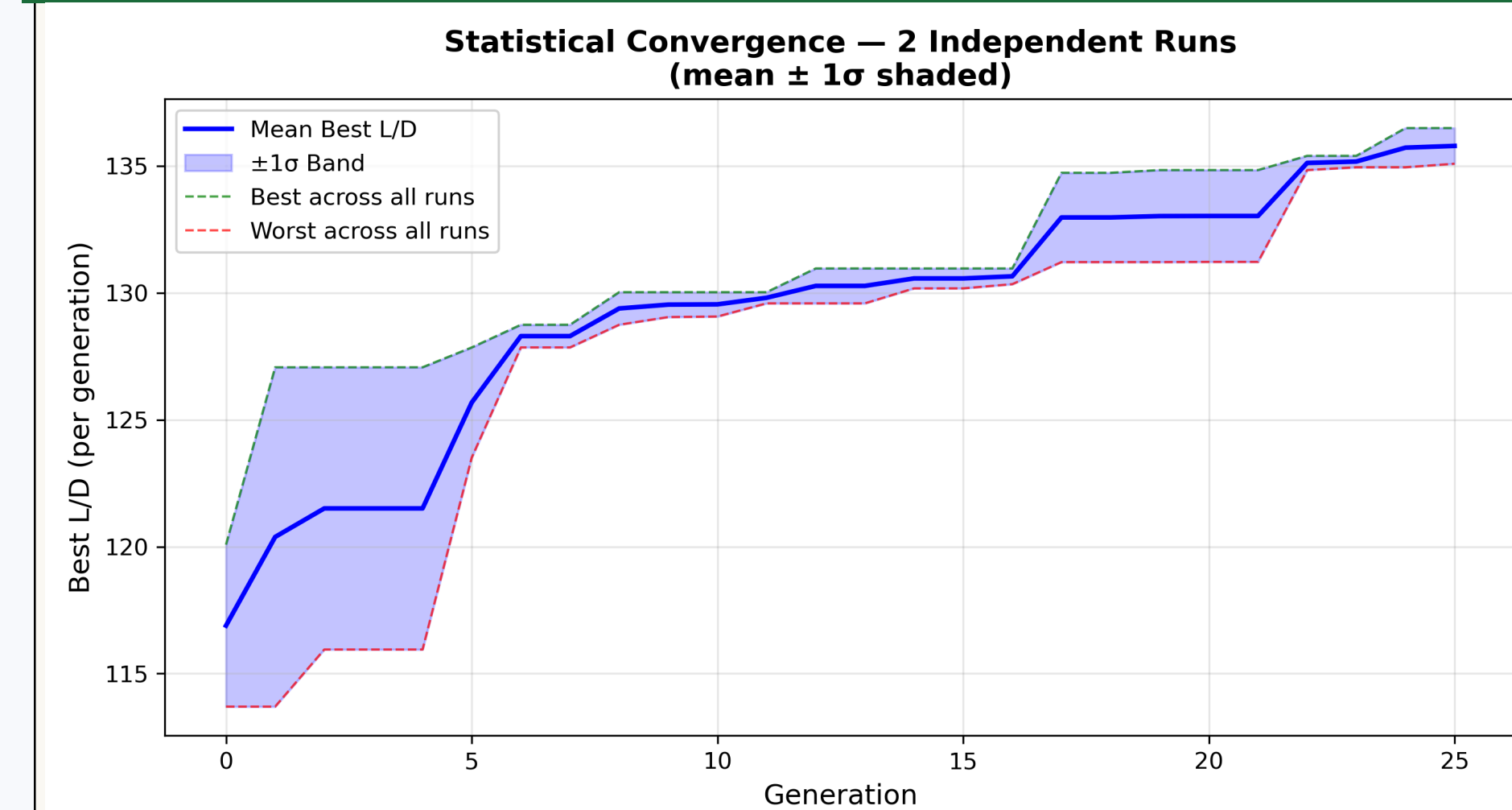


Figure 5. Statistical Convergence Plot. The progression of the best L/D ratio across generations for independent runs.

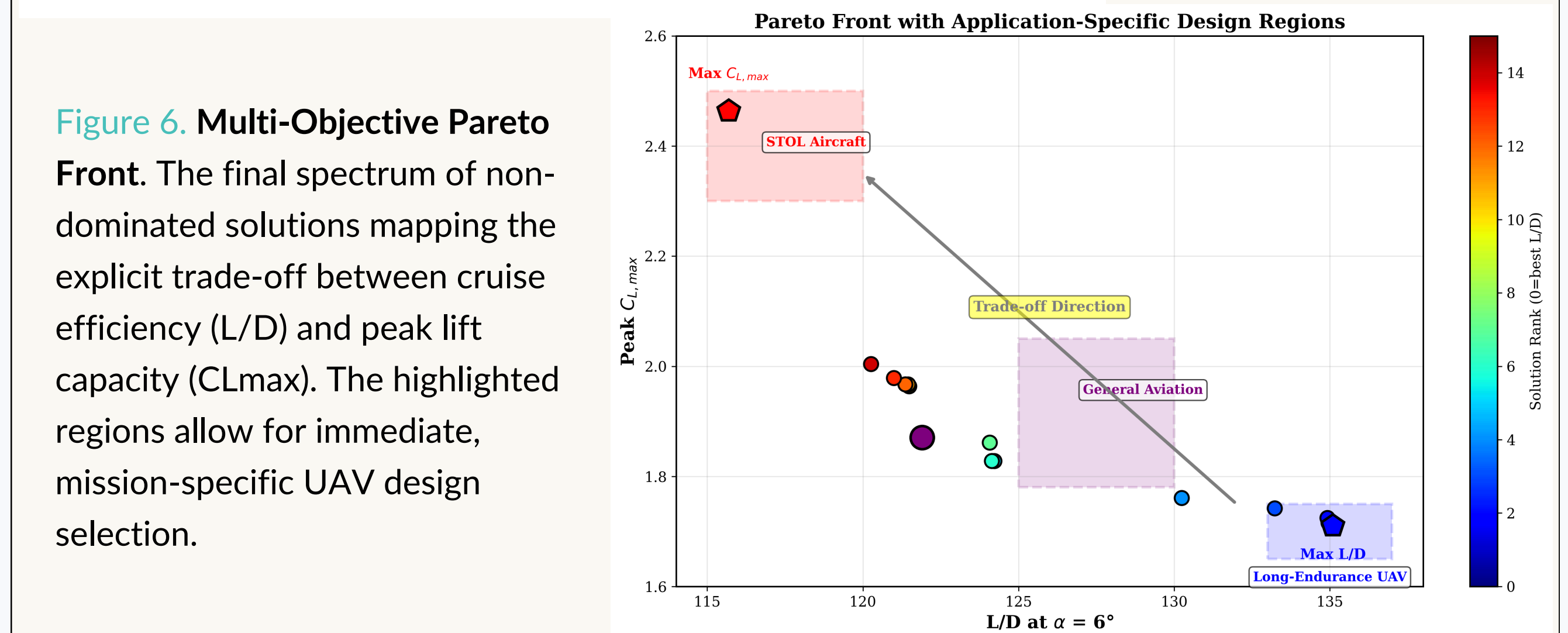


Figure 6. Multi-Objective Pareto Front. The final spectrum of non-dominated solutions mapping the explicit trade-off between cruise efficiency (L/D) and peak lift capacity (CL_{max}). The highlighted regions allow for immediate, mission-specific UAV design selection.

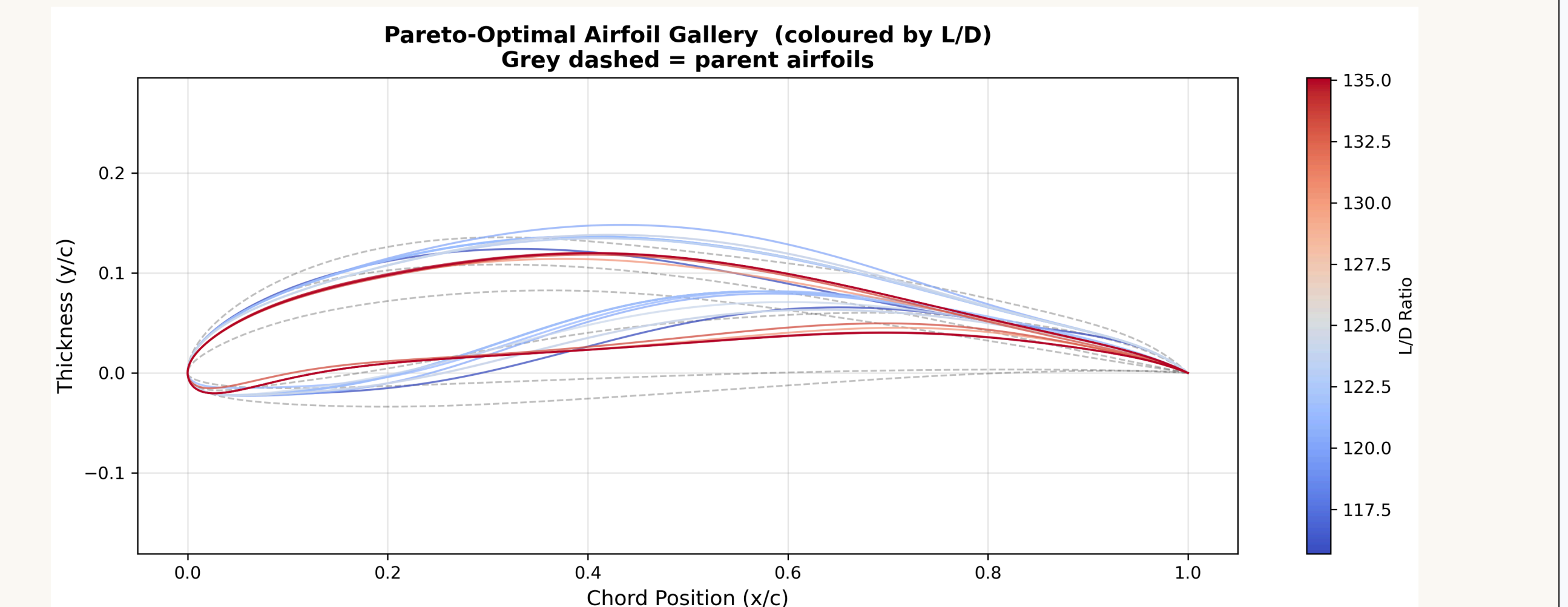


Figure 7. Pareto-Optimal Airfoil Gallery. A visual overlay of the non-dominated solution set generated by the NSGA-II algorithm, color-coded by peak L/D performance.

CONCLUSION & FUTURE WORK

- NSGA-II coupled with CST parameterisation provides an automated, effective framework for multi-objective airfoil design
 - Explicit Pareto mapping enables mission-specific airfoil selection across cruise efficiency and maximum lift trade-offs
 - Eliminates reliance on iterative trial-and-error in aerospace design
- Future Work**
- Extend framework to 3D wing planform optimisation incorporating induced drag and lifting-line theory
 - Validate 2D results using high-fidelity RANS CFD solvers (OpenFOAM)

REFERENCES

- [1] M. Drela, "XFOIL: An analysis and design system for low Reynolds number airfoils," in Conference on Low Reynolds Number Airfoil Aerodynamics, University of Notre Dame, 1989.
- [2] F.-A. Fortin, F.-M. De Rainville, M.-A. Gardner, M. Parizeau, and C. Gagné, "DEAP: Evolutionary Algorithms Made Easy," Journal of Machine Learning Research, vol. 13, pp. 2171–2175, 2012.
- [3] B. M. Kulfan, "Universal parametric geometry representation method," Journal of Aircraft, vol. 45, no. 1, pp. 142–158, 2008.